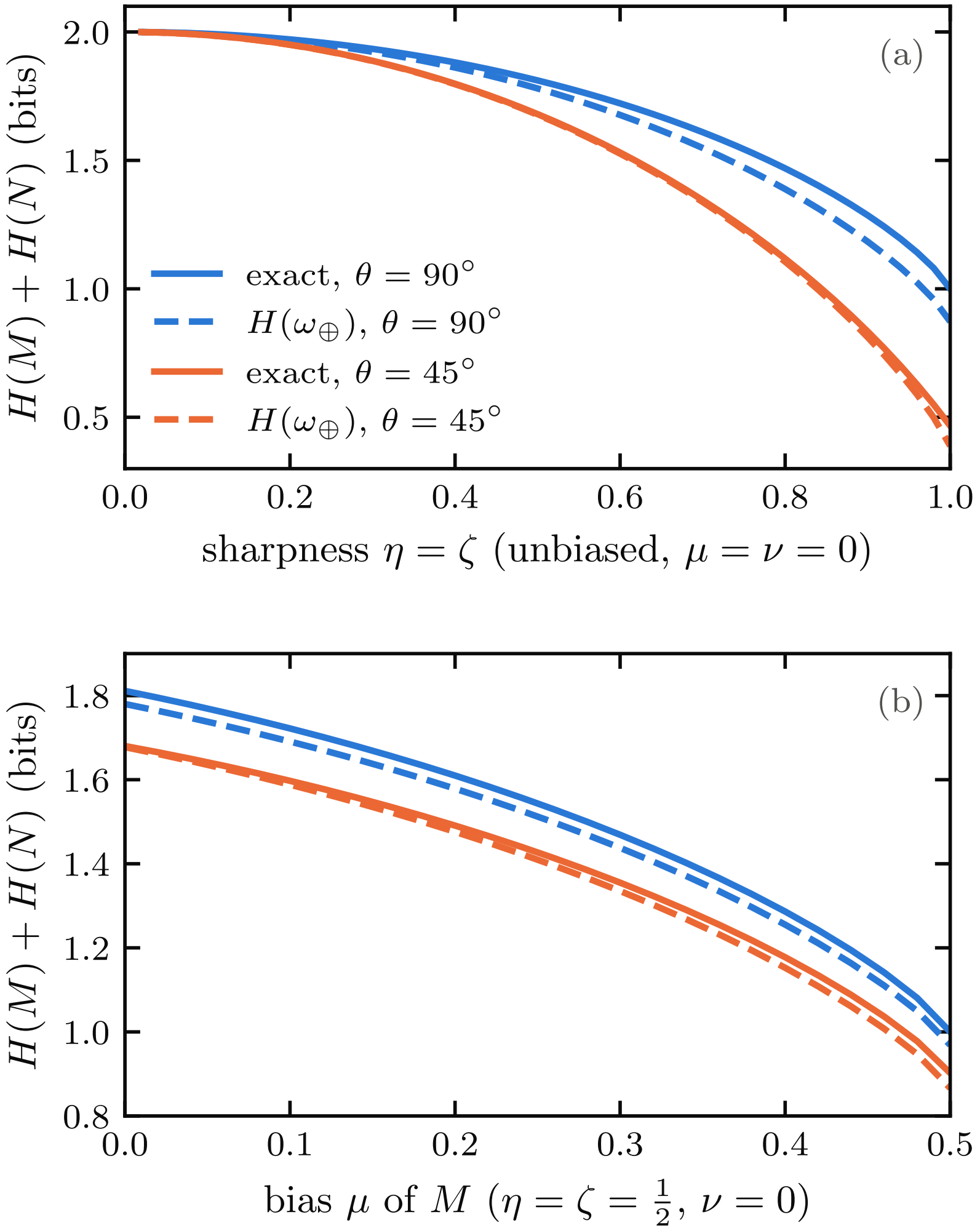

(a)
exact, $\theta = 90°$
$H(\omega_\oplus)$, $\theta = 90°$
exact, $\theta = 45°$
$H(\omega_\oplus)$, $\theta = 45°$
$H(M) + H(N)$ (bits)
2.0
1.5
1.0
0.5
0.0
0.2
0.4
0.6
0.8
1.0
sharpness $\eta = \zeta$ (unbiased, $\mu = \nu = 0$)
(b)
$H(M) + H(N)$ (bits)
1.8
1.6
1.4
1.2
1.0
0.8
0.0
0.1
0.2
0.3
0.4
0.5
bias μ of M ($\eta = \zeta = \frac{1}{2}$, $\nu = 0$)